

Oxidation-Mediated, Zwitterionic Polydopamine Coatings for Marine Antifouling Applications

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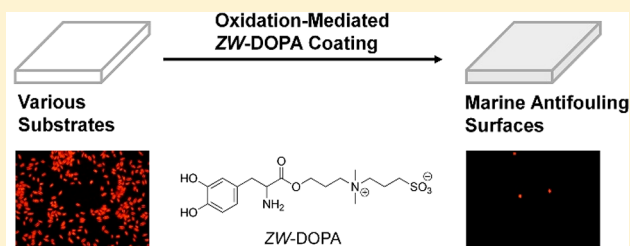
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Supporting Information

ABSTRACT: We synthesized a zwitterionic dopamine derivative (ZW-DOPA) containing both catechol and amine groups, and we demonstrated an excellent marine antifouling surface by controlling the oxidation of ZW-DOPA. The oxidation was mediated by the deprotonation of catechol or the addition of an oxidant (ammonium persulfate (AP) or sodium periodate (NaIO₄)). The oxidation and subsequent molecular transformation of ZW-DOPA was investigated over time by UV–vis spectroscopy. Among the different oxidation conditions tested, NaIO₄-induced ZW-DOPA coating was the most efficient and successfully formed on various substrates, such as titanium dioxide, stainless steel, and nylon. Compared with uncoated substrates, ZW-DOPA-coated substrates showed high resistance to marine diatom adhesion. Considering the ease of use and substrate independence of the ZW-DOPA coating, this method shows promise as a basis for inhibiting marine fouling on a variety of substrates used in the marine industry and aquatic environments.



INTRODUCTION

Marine fouling is the adhesion and accumulation of marine organisms, including bacteria, algae, and mollusks, onto surfaces submerged in seawater. Fouling increases hydrodynamic drag forces and reduces the speed of marine vessels, resulting in increased fuel consumption and subsequent emission of greater amounts of harmful compounds (e.g., carbon dioxide and sulfur dioxide) into the environment.^{1–3}

Marine fouling also leads to greater costs for hull repair, cleaning, and maintenance.^{1,4} One report estimated that marine fouling costs \$1 billion per year.⁵ Therefore, marine fouling is a crucial environmental and economic problem to be solved. Marine fouling generally occurs through the following steps:⁶ (i) the submerged surfaces are covered by organic macromolecules (e.g., proteins), (ii) microbial films are formed on the surfaces with the secretion of extracellular polymer, enabling the attachment of multicellular species (e.g., spores of macroalgae), and (iii) macrofoulers, such as macroalgae, barnacles, and mussels, finally attach on the top.

Biocides were initially used to inhibit marine fouling; however, their use is severely prohibited by environmental regulation. As approaches that are more eco-friendly, functional organic coatings that either release the attached foulants (foul-release) or prevent the attachment of foulants (antifouling) have received considerable attention.^{7,8} Poly(ethylene glycol) (PEG),^{9–12} zwitterionic polymers,^{13–16} polysaccharides,^{17–19} and poly(2-methyl-2-oxazolines)²⁰ have been

utilized to prepare marine antifouling coatings. In addition, amphiphilic polymers have been used to form antifouling coatings because the coexistence of hydrophobic and hydrophilic domains can lead to confusion in marine organisms during their attachment onto surfaces.^{21,22} The polymer-modified surfaces showed high resistance to the adsorption of proteins, bacteria, and microalgae, demonstrating their excellent marine antifouling effect.

When marine antifouling coatings are practically applied to solid surfaces, both the versatility and the simplicity of the coating process should be considered. In marine and aquatic industries, various materials including stainless steel (SS) and nylon are commonly used; therefore, antifouling materials with universal (i.e., substrate-independent) coating capabilities are required, which can widen their practical applicability. Most of the previous studies have focused on the development of excellent antifouling materials with one-step coating processes but were used only for limited substrates. As a substrate-independent marine antifouling coating, a very recent report utilized polydopamine as an adlayer to immobilize catechol-functionalized PEG via Fe³⁺-catechol coordination.²³ Although

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the coating was robust and versatile, it required three steps to achieve the marine antifouling coating. It would be more beneficial if universal marine antifouling coating could be prepared in a one-step process.

To design substrate-independent coating materials, the mussel-inspired approach has become popular.^{24,25} It was reported that 1,2-dihydroxybenzene (catechol) and the amine groups of 3-(3,4-dihydroxyphenyl)-L-alanine (L-DOPA) in the adhesive proteins of the mussel play important roles in the exceptional adhesion capability of mussels.²⁶ On the basis of these findings, the oxidative polymerization of 2-(3,4-dihydroxyphenyl)ethylamine (dopamine), which is a small organic molecule containing catechol and amine groups, has been widely used to prepare functional coatings on various organic/inorganic substrates²⁷ and even living cells.²⁸ L-DOPA derivatives have also been synthesized to develop one-step, facile functional coatings on any kind of surface. For example, we previously synthesized perfluorinated DOPA to prepare superhydrophobic, self-cleaning coatings on various substrates.²⁹ The control of oxidative polymerization of dopamine and DOPA derivatives was important to achieve high coating efficiency and to modulate surface properties.³⁰

In this work, we aimed to develop a one-step, versatile coating with excellent marine antifouling capability. We synthesized a zwitterionic DOPA (ZW-DOPA) containing both catechol and amine groups and successfully accomplished one-step, facile ZW-DOPA coatings on titanium dioxide (TiO₂) substrates by controlling the oxidation of ZW-DOPA with alkaline pH or the use of an oxidant. The optimized ZW-DOPA coating could be applied to SS and nylon, showing substrate-independent coating capability. We examined the marine antifouling capability of ZW-DOPA-coated substrates with *Amphora coffeaeformis* (*A. coffeaeformis*) and demonstrated that the surface coating was highly resistant to the adhesion of marine diatoms.

EXPERIMENTAL SECTION

Materials. 3-(3,4-Dihydroxyphenyl)-L-alanine (L-DOPA, >98.0%), 1,3-propane sultone (>99.0%), 1,8-diazabicyclo[5.4.0]-undec-7-ene (DBU, >98%), tetrabutylammonium fluoride solution (TBAF), trifluoroacetic acid (TFA, >99%), *N,N'*-dicyclohexylcarbodiimide (DCC, >98.0%), and 4-dimethylaminopyridine (DMAP, >99.0%) were purchased from Tokyo Chemical Industry Co., Ltd. Tetrahydrofuran (THF, anhydrous, ≥99.9%), acetonitrile (anhydrous, 99.8%), ammonium persulfate (AP, reagent grade, 98%), tris(hydroxymethyl)aminomethane hydrochloride (Trizma HCl, 99%), and tris(hydroxymethyl)aminomethane (Trizma base, 99%) were purchased from Sigma-Aldrich. Sodium periodate (NaIO₄, 99%, Acros Organics), 3-(dimethylamino)-1-propanol (99%, Alfa Aesar), di-*tert*-butyl dicarbonate (Boc₂O, 97+%, Alfa Aesar), *tert*-butyldimethylchlorosilane (TBDMSCl, 97%, Alfa Aesar), acetone (99.5%, Samchun Chemicals), ethanol (ACS grade, Honey well), dichloromethane (99.5%, Samchun Chemicals), methanol (ACS grade, Honey well), ethyl acetate (99.5%, Samchun Chemicals), hexane (95%, Samchun Chemicals), magnesium sulfate (MgSO₄, anhydrous, 99%, Daejung, South Korea), sodium chloride (99%, Daejung, South Korea), hydrogen chloride (35%, Daejung, South Korea), sodium hydroxide (97%, Daejung, South Korea), and LiChroprep RP18 (40–63 μm, Merck) were used as received. A titanium dioxide (TiO₂) substrate was prepared by the thermal evaporation of 100 nm of titanium onto a silicon wafer, and the top surface was oxidized naturally by exposing the wafer to air. Stainless steel (SS) foil and Nylon-6 film were purchased from Goodfellow Cambridge Ltd. Deionized (DI) water from a Human Ultra Pure System (Human Corp., South Korea) was used.

Synthesis of 3-((3-((2-Amino-3-(3,4-dihydroxyphenyl)propanoyloxy)propyl)dimethylammonio)propane-1-sulfonate, ZW-DOPA. L-DOPA (1) (2.0 g, 10.0 mmol) was dissolved in acetone (10 mL), and the L-DOPA solution was added to 1 M NaOH (11.0 mL, 11.0 mmol) and Boc₂O (2.4 g, 11.0 mmol) at 0 °C. The mixture solution was stirred for 4 h at room temperature. The solvent was removed by a rotary evaporator, and the resultant product was washed with 1 M HCl, water, and brine and then dried over MgSO₄. The product (Boc-DOPA) was concentrated in vacuo to give 2.8 g (94%), and the obtained Boc-DOPA was used without further purification. ¹H NMR (300 MHz, CDCl₃): δ 6.73–6.51 (m, 3H), 5.20 (d, *J* = 7.7 Hz, 1H), 4.44 (m, 1H), 2.98 (m, 2H), 1.39 (s, 9H). ¹³C NMR (75 MHz, CDCl₃): δ 175.86, 156.11, 143.91, 143.22, 127.96, 121.72, 116.30, 115.67, 80.97, 60.64, 37.19, 28.22.

To the acetonitrile solution (30.0 mL) of Boc-DOPA (2.6 g, 8.7 mmol) was added TBDMSCl (3.3 g, 21.8 mmol). The mixture was cooled to 0 °C, and DBU (3.3 mL, 21.8 mmol) was added dropwise over 10 min. The solution was stirred for 16 h at room temperature and then evaporated. The resultant product was washed with water and brine, and then dried over MgSO₄. The crude product was purified by silica-gel column chromatography (5% methanol/dichloromethane) to give 3.2 g (70%) of compound 2 as a yellow oil. ¹H NMR (300 MHz, CDCl₃): δ 6.75 (d, *J* = 8.0 Hz, 1H), 6.62 (m, 2H), 4.89 (d, *J* = 8.1 Hz, 1H), 4.52 (m, 1H), 3.00 (m, 2H), 1.42 (s, 9H), 0.97 (s, 18H), 0.18 (s, 12H). ¹³C NMR (75 MHz, CDCl₃): δ 175.75, 155.32, 146.67, 145.91, 129.16, 128.87, 122.32, 121.07, 80.02, 54.16, 37.04, 28.29, 25.92, 18.39, −4.12. EI-MS, *m/z*: calcd for C₂₆H₄₇NO₆Si₂Na [M + Na]⁺, 548.2840; found, [M + Na]⁺ 548.2854.

Compound 2 (3.0 g, 5.7 mmol) and 3-(dimethylamino)propan-1-ol (0.7 mL, 6.3 mmol) were dissolved in dichloromethane. DCC (1.3 g, 6.3 mmol) and DMAP (0.8 g, 6.3 mmol) were added to the solution. The reaction mixture was stirred for 18 h at room temperature, and the resulting suspension was filtered off. After washing with hexane, the residue was purified by silica-gel chromatography (7% methanol/dichloromethane) to give 1.7 g (50%) of compound 3 as a yellow oil. ¹H NMR (300 MHz, CDCl₃): δ 6.70 (d, *J* = 8.1 Hz, 1H), 6.55 (m, 2H), 4.97 (d, *J* = 7.8 Hz, 1H), 4.48 (m, 1H), 4.12 (m, 2H), 2.93 (d, *J* = 5.1 Hz, 2H), 2.27 (t, *J* = 7.1 Hz, 2H), 2.18 (s, 6H), 1.75 (m, 2H), 1.40 (s, 9H), 0.96 (s, 18H), 0.16 (s, 12H). ¹³C NMR (75 MHz, CDCl₃): δ 171.86, 155.04, 146.64, 145.88, 128.94, 122.21, 120.98, 79.68, 63.69, 56.06, 54.35, 45.40, 37.50, 28.30, 26.77, 25.90, 18.39, −4.11. EI-MS, *m/z*: calcd for C₃₁H₅₈N₂O₆Si₂Na [M + Na]⁺, 633.3731; found, [M + Na]⁺ 633.3443.

Compound 3 (1.5 g, 2.5 mmol) was dissolved in acetonitrile (20 mL), and 1,3-propane sultone (0.4 g, 2.9 mmol) was added to the solution. The solution mixture was stirred for 24 h and then evaporated. The crude product was purified by silica-gel chromatography (10–30% methanol/dichloromethane) to give 1.6 g (87%) of compound 4 as a white powder. ¹H NMR (300 MHz, CDCl₃): δ 6.76 (d, *J* = 8.1 Hz, 1H), 6.60 (m, 2H), 5.00 (d, *J* = 5.5 Hz, 1H), 4.46 (m, 1H), 4.32–4.05 (m, 2H), 3.75 (m, 2H), 3.45 (m, 2H), 3.18 (s, 6H), 3.00–2.82 (m, 4H), 2.22 (m, 4H), 1.37 (s, 9H), 0.98 (s, 18H), 0.19 (s, 12H). ¹³C NMR (75 MHz, CDCl₃): δ 172.22, 155.42, 146.77, 145.97, 128.81, 122.12, 121.97, 121.20, 79.86, 63.24, 61.43, 54.77, 50.92, 47.70, 36.74, 28.34, 25.92, 22.55, 19.36, 18.41, −4.09. EI-MS, *m/z*: calcd for C₃₄H₆₄N₂O₉SSi₂Na [M + Na]⁺, 755.3769; found, [M + Na]⁺ 755.3794.

One molar TBAF (4.8 mL, 4.8 mmol) was added to the solution of compound 4 (1.4 g, 1.9 mmol) in THF (20 mL) at 0 °C. The resulting mixture was stirred at 0 °C for 3 h and then evaporated. The crude product was dissolved in dichloromethane (20 mL), and TFA (1.76 mL, 22.9 mmol) was added to the solution at 0 °C. The reaction mixture was stirred for 3 h at room temperature and then evaporated. The crude product was purified by C-18 reversed phase silica-gel column chromatography (10% H₂O/methanol) and then freeze-dried to give 0.52 g (68%) of ZW-DOPA as a yellow powder. ZW-DOPA was divided into several vials and stored in a freezer prior to use. ¹H NMR (300 MHz, DMSO): δ 6.71 (d, *J* = 8.0 Hz, 1H), 6.64 (d, *J* = 2.0 Hz, 1H), 6.46 (dd, *J* = 8.0, 2.0 Hz, 1H), 4.17 (m, 2H), 4.03

Scheme 1. Synthetic Procedure of 3-((3-((2-Amino-3-(3,4-dihydroxyphenyl)propanoyl)oxy)propyl)dimethylammonio)propane-1-sulfonate, ZW-DOPA

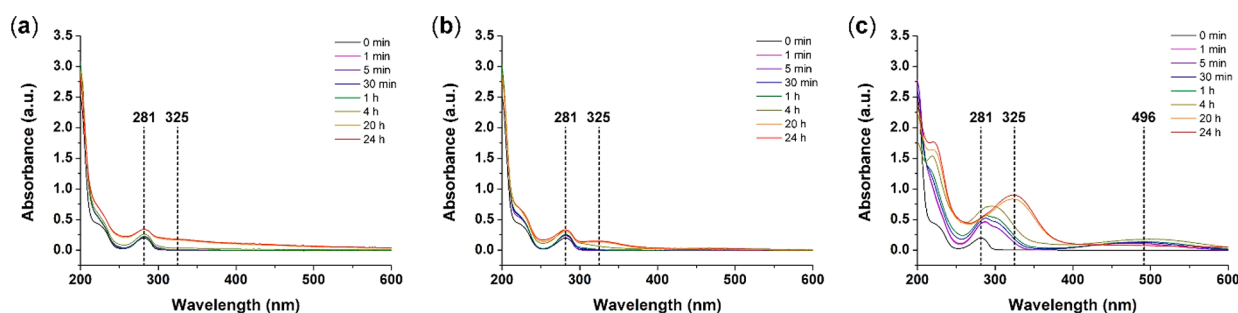
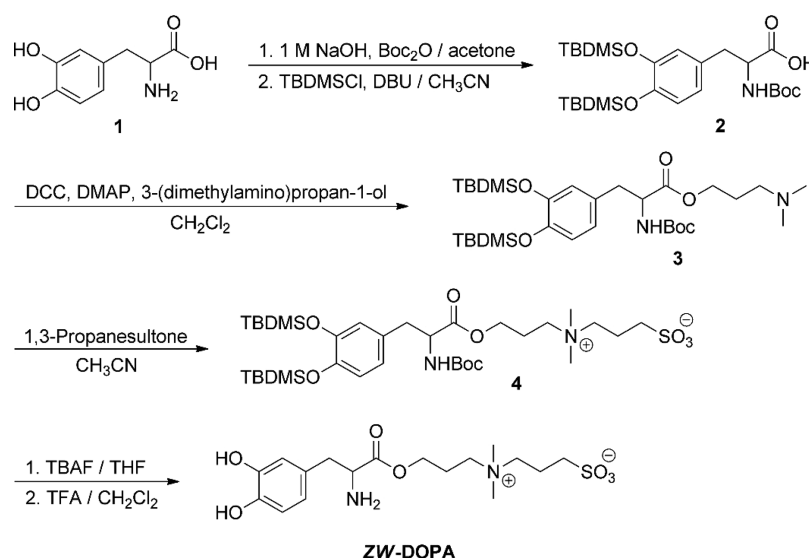


Figure 1. Changes in UV-vis spectra of ZW-DOPA solutions with different oxidation conditions: (a) Tris buffer (pH 8.5), (b) ammonium persulfate (AP), and (c) sodium periodate (NaIO_4). The spectrum at 0 min indicates the absorbance from the solution containing only ZW-DOPA.

(m, 1H), 3.38 (m, 2H), 3.12 (m, 2H), 3.04–2.81 (m, 2H), 2.98 (s, 6H), 2.52 (m, 2H), 1.94 (m, 4H). ^{13}C NMR (75 MHz, DMSO): δ 169.31, 145.69, 145.03, 125.64, 120.65, 117.17, 116.24, 62.75, 62.31, 60.12, 54.05, 50.74, 47.96, 36.08, 21.78, 19.09. EI-MS, m/z : calcd for $\text{C}_{17}\text{H}_{29}\text{N}_2\text{O}_7\text{S}$ $[\text{M} + \text{H}]^+$, 405.1695; found, $[\text{M} + \text{H}]^+$ 405.1696.

ZW-DOPA Coating. Prior to the coating, a substrate was cleaned with acetone and ethanol under sonication for 5 min each and dried under a stream of argon (Ar) gas. ZW-DOPA was dissolved in Tris buffer (pH 8.5, 50 mM) with the concentration of 2 mg/mL. When the surface coating was carried out with AP or NaIO_4 , ZW-DOPA (2 mg/mL) and an oxidant (AP or NaIO_4) were dissolved in DI water with the molar ratio of 2:1. The prepared solution was poured into a vial containing the cleaned substrate. After 24 h, the substrate was washed with DI water and then dried under a stream of Ar gas.

Diatom Adhesion Test. *Amphora coffeaeformis* (*A. coffeaeformis*) was obtained from the Korea Marine Microalgae Culture Center (KMMCC) and cultured in 100 mL of f/2 medium at 18 °C. A log-phase suspension of cells was diluted with f/2 medium to give a cell suspension with a chlorophyll *a* content of $\sim 3.0 \mu\text{g/mL}$. A 13 mL aliquot of cell suspension was added to the petri dishes containing the solid substrates. After 24 h at 20 °C, the substrates were gently washed in filtered seawater to remove unattached cells. The attached cells were characterized and counted using an optical microscope (LV100ND, Nikon, Japan) equipped with a fluorescence imaging system.

Characterizations. Synthesized compounds were characterized by nuclear magnetic resonance spectroscopy (NMR, Bruker), operated at the ultrashield of 300 MHz, and high resolution mass spectroscopy (MR-MS, Bruker Daltonics). UV-vis spectra were obtained using an UV-vis spectrophotometer (Agilent 8453, U.S.A.).

The film thicknesses were measured with a spectroscopic ellipsometer at a 70° angle of incidence using wavelengths (λ) from 400 to 700 nm (Elli-SE, Ellipso Technology, Suwon, South Korea). Contact angle measurements were performed using a Phoenix 10 apparatus (Surface Electro Optics Co. Ltd., Korea) equipped with a video camera. The static contact angles of water drops of $\sim 5 \mu\text{L}$ were measured at three different locations on each sample, and the average values were reported in this paper. The X-ray photoelectron spectroscopy (XPS) study was performed with a VG-Scientific MultiLab 2000 spectrometer (Thermo Scientific) with a monochromatized Al $K\alpha$ X-ray source (1486.6 eV). Atomic force microscopy (AFM) images were obtained in tapping mode on a NX-10 scanning probe microscope (Park Systems).

RESULTS AND DISCUSSION

Strong hydration is critical for antifouling coatings.³¹ PEG, the most common antifouling material, interacts with water molecules via hydrogen bonding, whereas zwitterionic compounds bind to water by electrostatic interactions using both negative and positive charges in the compounds, leading to stronger hydration. In addition, zwitterionic materials have shown ultralow protein fouling.^{32,33} Therefore, we designed a zwitterionic DOPA derivative. Briefly, L-DOPA was coupled with 3-(dimethylamino)-1-propanol via esterification with protection/deprotection of the hydroxy and amine groups (Scheme 1). It was reported that the introduction of a chemical moiety into L-DOPA was compatible with the oxidative polymerization.²⁹ The successful syntheses of the

intermediates and the final product, ZW-DOPA, were confirmed by ^1H NMR, ^{13}C NMR, and mass spectrometry (See [Experimental Section](#) for the details).

In oxidative polymerization of dopamine, the catechol group is oxidized into *o*-quinone, followed by intramolecular cyclization via Michael-type conjugation addition, further oxidation into dopaminechrome, and then isomerization into 5,6-dihydroxyindole. It is notable that the 5,6-dihydroxyindole moiety is highly involved in the polymerization process through both covalent and noncovalent interactions ([Scheme S1](#)).^{34,35} In this study, the oxidation of ZW-DOPA was mediated by alkaline pH or an oxidant. We prepared three ZW-DOPA solutions: ZW-DOPA was dissolved in Tris buffer (pH 8.5, 50 mM) to a concentration of 2 mg/mL. When an oxidant was used, ZW-DOPA (2 mg/mL) and an oxidant (ammonium persulfate (AP) or sodium periodate (NaIO_4)) were dissolved in DI water with the molar ratio of 2:1. We investigated the oxidation process of ZW-DOPA by analyzing the diluted ZW-DOPA solutions with UV-vis spectroscopy ([Figure 1](#)). The UV-vis spectrum of ZW-DOPA showed a characteristic peak at 281 nm, representing the symmetry-forbidden transition (L_a-L_b) of the catechol group in ZW-DOPA.³⁶ The spectrum changed over time, and changes were more evident when a relatively strong oxidant, NaIO_4 , was used. Upon oxidation, the intensity of the peak at 281 nm decreased. A new peak appeared as a shoulder over the peak at 281 nm and was clearly observed at 325 nm after 20–24 h. The peak at 325 nm indicated the oxidation of catechol into *o*-quinone.³⁷ A weak and broad peak was also observed at 496 nm, indicating the formation of dopaminechrome. The UV-vis analysis confirmed that ZW-DOPA followed the procedure applied for the oxidative polymerization of dopamine.

It is noteworthy, however, that the oxidation of ZW-DOPA is quite slow compared to that of a perfluorinated DOPA derivative, which was synthesized via esterification of L-DOPA and perfluoroalcohol in our previous work.²⁹ The oxidation and subsequent reactions of perfluorinated DOPA with NaIO_4 proceeded within 2 h, whereas the oxidative reactions of ZW-DOPA required 20–24 h. The slow oxidation of ZW-DOPA may be related to the strong interaction between ZW-DOPA and water molecules in the aqueous solutions. ZW-DOPA exhibits an extremely high dipole moment due to a sulfobetaine group,^{38,39} and the molecules in solid phase associate with one another by electrostatic interactions; however, the electrostatic attractions should be reduced in water, a solvent with high dielectric constant ($\epsilon \approx 80$), resulting in strong hydration of ZW-DOPA molecules⁴⁰ and the formation of hydrogen bonds between water and catechol groups. Such strong interactions would increase the redox potential of catechol groups, slowing the oxidation of ZW-DOPA.^{41,42} For AP-mediated oxidation of ZW-DOPA, the slow decomposition of persulfate into the strongly oxidizing sulfate radical anion, which actually oxidizes catechol groups, gave rise to the slower oxidation than NaIO_4 -induced oxidation.³⁰

To examine the coating capability of ZW-DOPA under different oxidation conditions (alkaline pH or the use of an oxidant), we employed three ZW-DOPA coating solutions as previously mentioned. We coated Ti/TiO_2 substrates with ZW-DOPA by immersing substrates into the various prepared solutions for 24 h. While the 1.1 nm thick film was formed on the substrate in the absence of oxidants, thicker films were

formed with AP and NaIO_4 , producing 3.7 and 39.5 nm thick films, respectively ([Figure 2](#)).

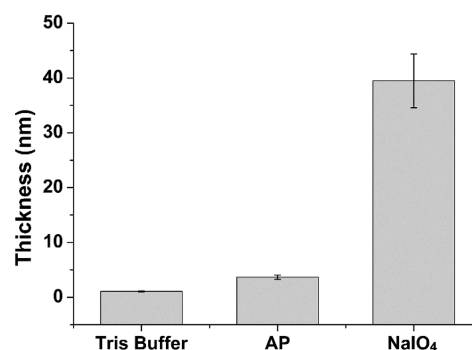


Figure 2. ZW-DOPA film thickness formed on Ti/TiO_2 substrates with different oxidation conditions.

The low coating efficiency under alkaline conditions (with no additional oxidant) is consistent with the previous report in which the Ti/TiO_2 substrate was not coated with L-DOPA in bicine buffer (10 mM, pH 8.5).⁴³ The extremely thin film formed under alkaline conditions may indicate that the film would be formed as a monolayer-level via self-assembly, rather than oxidative polymerization. This formation could be supported by the extremely slow oxidation of ZW-DOPA in Tris buffer: the oxidation process was negligible, as observed in the UV-vis spectrum during the 24 h reaction ([Figure 1a](#)).

The formation of ZW-DOPA films on Ti/TiO_2 substrates was further investigated with water contact angle goniometry, X-ray photoelectron spectroscopy (XPS), and atomic force microscopy (AFM). The static water contact angle on the TiO_2 substrate decreased from 47.0° to 37.7° after ZW-DOPA coating in the absence of an additional oxidant. With oxidant-induced polymerization, the ZW-DOPA-coated substrate became more hydrophilic, with the water contact angle below 10° . Such a reduction in the water contact angle indicates multilayered ZW-DOPA coatings on the surfaces ([Figure 3](#)).

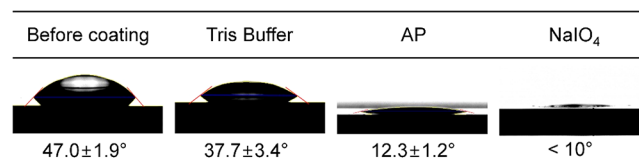


Figure 3. Static water contact angles on Ti/TiO_2 substrates before and after ZW-DOPA coating with different oxidation conditions.

XPS analysis provided more direct evidence for ZW-DOPA coatings on surfaces. With the ZW-DOPA coating, the new peaks appeared at 400.8 (N 1s), 230.8 (S 2s), and 166.8 (S 2p) eV, and the quantitative atomic composition of C 1s increased with a concurrent decrease of Ti 2p ([Figure 4](#) and [Table S1](#)). AFM analysis also revealed the formation of ZW-DOPA coatings. The pH- or AP-induced surface coating was uniform, with surface topography similar to that of bare substrate. The thicker ZW-DOPA coating with NaIO_4 showed rough surface features. The root-mean-square roughness (R_q) increased from 3.41 to 6.70 nm after the coating ([Figure 5](#)).

To evaluate the marine antifouling property of the oxidation-mediated ZW-DOPA films, we examined the adhesion

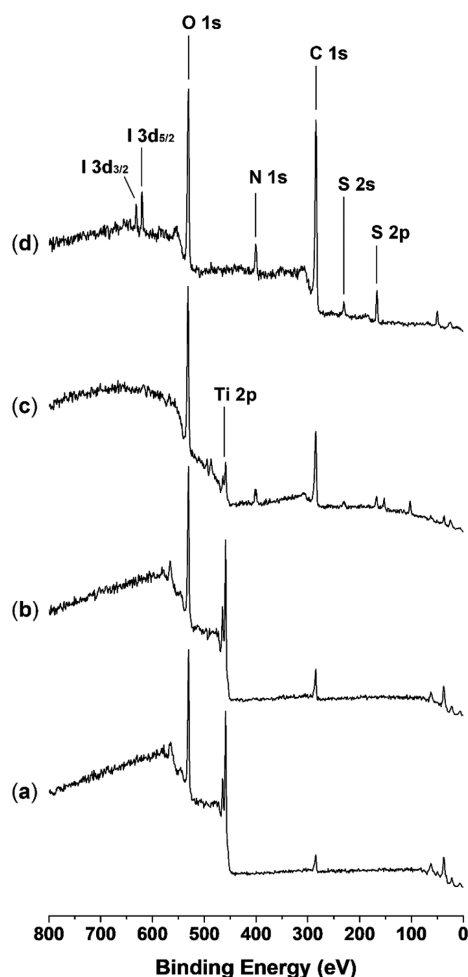


Figure 4. Survey XPS spectra of (a) bare and (b–d) ZW-DOPA-coated Ti/TiO₂ substrates with different oxidation conditions: (b) Tris buffer (pH 8.5), (c) AP, and (d) NaIO₄.

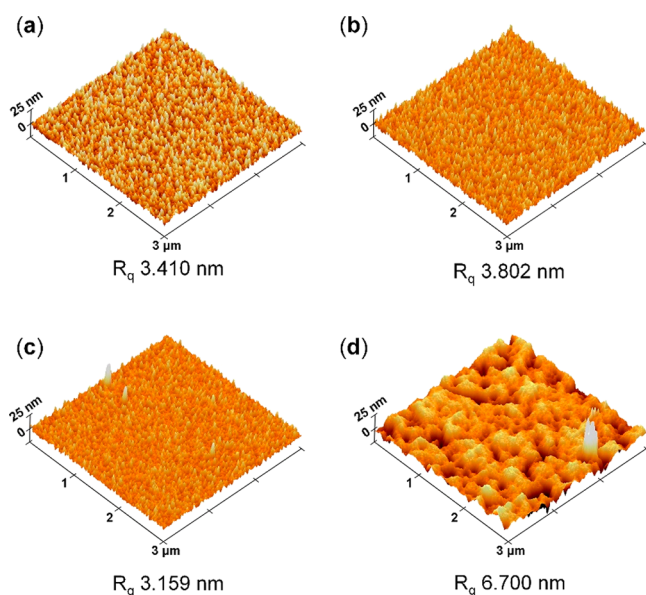


Figure 5. AFM images of (a) bare and ZW-DOPA-coated Ti/TiO₂ substrates with different oxidation conditions: (b) Tris buffer (pH 8.5), (c) AP, and (d) NaIO₄.

behavior of marine diatoms. *A. coffeaeformis*, commonly used in laboratory assays, was selected as a model species.^{44,45} Diatoms were cultured in f/2 medium for 10 days and used for the adhesion assay (with chlorophyll a content of 3 μg/mL). After a 24 h incubation with bare and ZW-DOPA-coated substrates, the attached diatom density on different substrates was quantitatively analyzed. Figure 6 shows that the attached

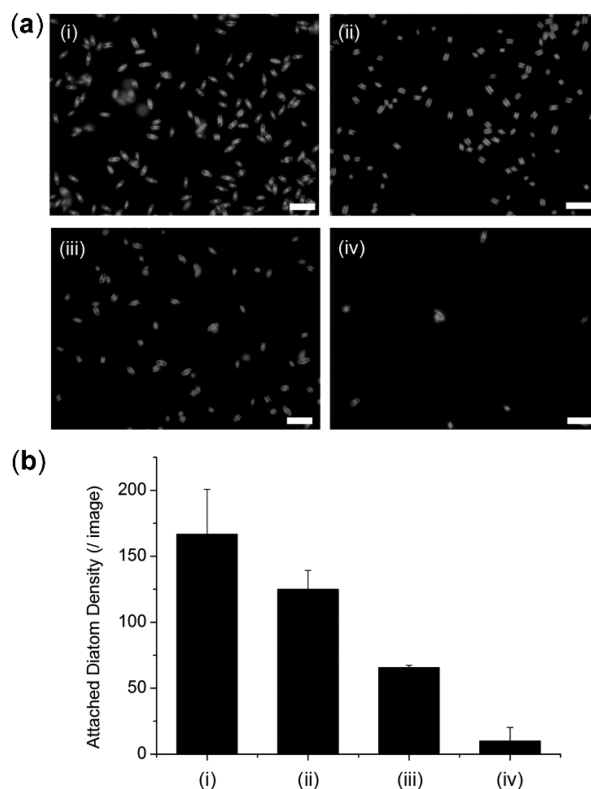


Figure 6. (a) Representative fluorescence images and (b) quantification of *A. coffeaeformis* diatoms attached on (i) bare and ZW-DOPA-coated Ti/TiO₂ substrates with different oxidation conditions: (ii) Tris buffer (pH 8.5), (iii) AP, and (iv) NaIO₄. Each point represents the mean of 60 counts on three replicate samples. Error bars display the 95% confidence limits. Scale bars represent 50 μm.

diatom density decreased with ZW-DOPA coatings. Depending on the oxidation conditions, the adhesion density was reduced from 165 to 125, 66, and 10 diatoms/image for Tris buffer, AP, and NaIO₄ solutions, respectively. The coating conditions with NaIO₄ were the most effective in suppressing diatom adhesion. Such a marine antifouling capability of the ZW-DOPA coating likely resulted from the greatly enhanced surface hydrophilicity after the coating. It was reported that zwitterions have shown ultralow fouling properties due to the formation of a strong hydration layer,^{46,47} making the replacement of surface-bound water molecules with foulants (e.g., diatoms) enthalpically unfavorable.⁴⁸

As described above, we mediated the oxidation of ZW-DOPA and its coatings with alkaline pH and oxidants. Both high coating efficiency and excellent marine antifouling properties were accomplished with NaIO₄-induced ZW-DOPA films. To examine the substrate-independent coating capability of ZW-DOPA, we applied the NaIO₄-induced ZW-DOPA coatings to SS and nylon substrates. SS is the major component of the bodies of marine vessels to which marine

organisms severely adhere.¹² Nylon is used in ropes, stern tubes, rudder stocks of ships, and the preparation of aquaculture instruments. SS and nylon substrates developed a brown coloration after being coated with ZW-DOPA (Figure S1).

Figure S2 shows that the surface hydrophilicity of both SS and nylon substrates was significantly enhanced after ZW-DOPA coatings. The static water contact angle on SS was reduced from 60.7° to below 10°, and it decreased from 70.4° to 20.7° in the case of nylon. The formation of ZW-DOPA coatings on SS and nylon was further confirmed by XPS and AFM analysis. Upon coating the SS substrate with ZW-DOPA, the intrinsic Fe peak disappeared (13.95 to 0.00% in atomic composition) whereas the characteristic peaks including N 1s, S 2s, and S 2p clearly appeared. For the nylon substrate, new peaks of S 2s and S 2p appeared with concurrent decrease of the N 1s peak (11.14 to 6.78% in atomic composition) after ZW-DOPA coating (Figure S3 and Table S2). AFM images also supported the formation of the ZW-DOPA coating on SS and nylon substrates. The ZW-DOPA coating made the SS substrate slightly rougher, with an R_q value of 6.190 nm. Nylon exhibited decreased R_q from 6.034 to 3.353 nm after being coated with ZW-DOPA (Figure S4).

After confirmation of successful ZW-DOPA coatings on SS and nylon substrates, we performed diatom adhesion assays for the substrates. As shown in Figure S5, the ZW-DOPA coating completely inhibited diatom adhesion on the SS substrate: the attached diatom density on SS was significantly reduced from 288 to 3 diatoms/image after ZW-DOPA coating. ZW-DOPA-coated nylon, however, showed a reduced antifouling effect. The diatom adhesion density on nylon was reduced by 42% upon ZW-DOPA coating. The difference between the two ZW-DOPA-coated substrates in terms of their antifouling effects may have resulted from the relatively decreased hydrophilic character of ZW-DOPA-coated nylon compared with the ZW-DOPA-coated SS substrate, which was superhydrophilic, enabling strong hydration layers to be formed on the surface. It has been known that diatom adhesion can be reduced by enhancement of surface hydrophilicity.^{49–51} Although it demonstrated a relatively small effect on nylon, the ZW-DOPA coating exhibited its marine antifouling capabilities for various substrates via a simple, one-step process.

To widen applications from the practical point of view, the long-term stability of ZW-DOPA coatings should be examined because marine vessels are generally immersed in sea for a long time. The ZW-DOPA-coated SS substrates were immersed into seawater. After 28 days, the substrates were taken out and diatom adhesion assay was performed with the seawater-treated substrates. Compared with a bare substrate, the diatom adhesion was reduced by 52% on the ZW-DOPA-coated SS substrate (Figure S6). Although lowered, the marine antifouling effect of the ZW-DOPA coating was retained even after 28 days.

CONCLUSIONS

In this study, we synthesized a zwitterionic DOPA derivative, ZW-DOPA, and demonstrated the one-step preparation of ultralow marine fouling coatings on various substrates, such as TiO₂, SS, and nylon, through the control of ZW-DOPA oxidation. The NaIO₄-mediated coating was optimal for both high coating efficiency and marine antifouling capability. Although the marine antifouling effect of the ZW-DOPA coating was slightly reduced after a 28 day treatment in

seawater, we believe that the robustness of the ZW-DOPA coating can be improved by repeated deposition of ZW-DOPA with a spray-based coating system. Given the ease of use and versatility of the coating, the current approach can potentially be employed for surface modification of various materials and equipment used in the marine industry. This coating can also serve as a means of controlling surface hydrophilicity in various research areas, such as materials science, chemistry, and bioengineering.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.langmuir.8b03454.

Schematic procedure for the oxidative polymerization of ZW-DOPA, atomic composition (%) of bare and ZW-DOPA-coated Ti/TiO₂ substrates with different oxidation conditions, optical photograph of the substrates before and after ZW-DOPA coatings with NaIO₄, static water contact angles on SS and nylon substrates before and after ZW-DOPA coating with NaIO₄, XPS spectra of bare and ZW-DOPA-coated substrates with NaIO₄ for SS and nylon, atomic composition (%) of bare and ZW-DOPA-coated substrates with NaIO₄ for SS and nylon, AFM images of bare and ZW-DOPA-coated substrates with NaIO₄ for SS and nylon, representative fluorescence images and quantification of *A. coffeaeformis* diatoms attached on SS and nylon substrates before and after ZW-DOPA coating, quantification of *A. coffeaeformis* diatoms attached on the ZW-DOPA-coated SS substrates after immersion in seawater for 28 days (PDF)

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Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

ZW-DOPA, 3-((3-((2-amino-3-(3,4-dihydroxyphenyl)propanoyl)oxy)propyl)dimethylammonio)propane-1-sulfonate; AP, ammonium persulfate; NaIO₄, sodium periodate; PEG, poly(ethylene glycol); SS, stainless steel; L-DOPA, 3-

(3,4-dihydroxyphenyl)-L-alanine; Boc_2O , di-*tert*-butyl dicarbonate; TBDMSCl, *tert*-butyldimethylchlorosilane; DBU, 1,8-diazabicyclo[5.4.0]-undec-7-ene; DCC, *N,N'*-dicyclohexylcarbodiimide; DMAP, 4-dimethylaminopyridine; TBAF, tetrabutylammonium fluoride; THF, tetrahydrofuran; TFA, trifluoroacetic acid; TiO_2 , titanium dioxide; NMR, nuclear magnetic resonance; HR-MS, high resolution mass spectroscopy; XPS, X-ray photoelectron spectroscopy; AFM, atomic force microscopy.

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